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Micromirror device and optical scanning device

Abstract

The micromirror device includes: a mirror portion; a first support portion that swingably supports the mirror portion around a first axis; a pair of movable frames that face each other across the first axis; a second support portion that swingably supports a movable portion around a second axis; a driving portion that surrounds the movable portion and has a gap with the second support portion on the second axis; a coupling portion that couples the second support portion and the driving portion; and a fixed frame, in which, in a state where the mirror portion rotates around the first axis and an absolute value of a rotation angle is larger than 0 degrees, assuming that, in a plane orthogonal to the first axis and including the second axis, a distance between an intersection between the second axis and a straight line located on a surface of the second support portion and including each end point of the second support portion and an end part of the second support portion on a mirror portion side in a stationary state is denoted by A, and a total length of the second support portion in a direction of the second axis is denoted by L, a relationship of $\frac{2}{3} < A/L$ is satisfied.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
12019235	12/2023	Naono	N/A	N/A
2008/0285103	12/2007	Mizumoto	359/199.1	G02B 26/0858
2010/0253991	12/2009	Yamada	359/208.2	G02B 17/023
2011/0141538	12/2010	Mizumoto	N/A	N/A
2011/0181933	12/2010	Kubo et al.	N/A	N/A
2017/0210278	12/2016	Matsuno	N/A	N/A
2017/0343795	12/2016	Grutzeck et al.	N/A	N/A
2019/0265462	12/2018	Yamada	N/A	G02B 26/10

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3872556	12/2020	EP	G02B 26/0858
4 249 985	12/2022	EP	N/A
2017-132281	12/2016	JP	N/A
WO 2009/041342	12/2008	WO	N/A
WO 2010/021216	12/2009	WO	N/A
WO 2010/035759	12/2009	WO	N/A
WO 2020/085063	12/2019	WO	N/A

OTHER PUBLICATIONS

Extended European Search Report for European Application No. 22192004.4, dated Feb. 7, 2023.
cited by applicant

European Communication pursuant to Article 94(3) EPC for European Application No.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. 2021-146275 filed on Sep. 8, 2021. The above application is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND

1. Technical Field

(2) The technique of the present disclosure relates to a micromirror device and an optical scanning device.

2. Description of the Related Art

(3) A micromirror device (also referred to as a microscanner) is known as one of micro electro mechanical systems (MEMS) devices manufactured using the silicon (Si) microfabrication technique. Since the micromirror device is small and has low power consumption, it is expected to have a wide range of applications in laser displays, laser projectors, optical coherence tomography, and the like.

(4) There are various drive methods for the micromirror device, and a piezoelectric drive method using deformation of a piezoelectric body is promising since the generated torque is higher than that in other methods and a high scan angle can be obtained. In particular, in a case where a high scan angle is required, such as in a laser display, a higher scan angle can be obtained by resonantly driving the micromirror device of the piezoelectric drive method.

(5) A general micromirror device used in a laser display comprises a mirror portion and a piezoelectric actuator (see, for example, JP2017-132281A and WO2009/041342A). The mirror portion is swingable around a first axis and a second axis that are orthogonal to each other. The actuator allows the mirror portion to swing around the first axis and the second axis according to the driving voltage supplied from the outside. The above-described scan angle corresponds to a deflection angle of the mirror portion.

(6) As performance indicators of the laser display, resolution and viewing angle are mentioned. The resolution and viewing angle are related to a swing frequency and a deflection angle of the mirror portion of the micromirror device. For example, in a laser display of a Lissajous scanning method, two-dimensional optical scanning is performed by allowing the mirror portion to swing sequentially at two different frequencies around the first axis and the second axis. The larger the deflection angle of the mirror portion, the larger the scanning area of light, and the larger the image can be displayed with the shorter optical path length.

SUMMARY

(7) Generally, in a case where the micromirror device is driven resonantly, a crosstalk between axes is generated in which the swing of the mirror portion around one of the first axis and the second axis affects the swing of the mirror portion around the other axis. Specifically, according to the size

of a rotation angle around the one axis, a resonance frequency around the other axis shifts. As a result, the stability of the two-dimensional optical scanning is significantly reduced.

(8) For example, in a case where a driving frequency around the one axis is swept to bring the driving frequency closer to the resonance frequency to increase a deflection angle of the mirror portion, a deflection angle of the mirror portion around the other axis is greatly changed by the shift of the resonance frequency caused by the crosstalk between axes. In this case, a change in the deflection angle of the mirror portion around the other axis causes a shift of the resonance frequency around the axis on a side where the driving frequency is swept. That is, the crosstalk between axes causes a kind of feedback phenomenon.

(9) In this way, in a case where a shift amount of the resonance frequency by the crosstalk between axes is large, even though the driving frequency is swept, there are problems that the driving frequency cannot be adjusted to the resonance frequency due to the feedback phenomenon, and that the deflection angle of the mirror portion cannot be increased.

(10) An object of the present disclosure is to provide a micromirror device and an optical scanning device capable of suppressing a shift of a resonance frequency by a crosstalk between axes.

(11) In order to achieve the object, a micromirror device of the present disclosure comprises: a mirror portion on which a reflecting surface for reflecting incident light is formed; a first support portion that is connected to the mirror portion on a first axis located in a plane including the reflecting surface of the mirror portion in a stationary state, and that swingably supports the mirror portion around the first axis; a pair of movable frames that are connected to the first support portion and face each other across the first axis; a second support portion that is connected to the movable frame on a second axis which is located in the plane including the reflecting surface of the mirror portion in the stationary state and is orthogonal to the first axis, and that swingably supports a movable portion including the mirror portion, the first support portion, and the movable frame around the second axis; a driving portion that surrounds the movable portion and has a gap with the second support portion on the second axis; a coupling portion that couples the second support portion and the driving portion; and a fixed frame that is connected to the driving portion and surrounds the driving portion, in which, in a state where the mirror portion rotates around the first axis and an absolute value of a rotation angle is larger than 0 degrees, assuming that, in a plane orthogonal to the first axis and including the second axis, a distance between an intersection between the second axis and a straight line located on a surface of the second support portion and including each end point of the second support portion and an end part of the second support portion on a mirror portion side in the stationary state is denoted by A, and a total length of the second support portion in a direction of the second axis is denoted by L, a relationship of $\frac{2}{3} < A/L$ is satisfied.

(12) It is preferable that the driving portion has a piezoelectric element.

(13) It is preferable that the driving portion includes a pair of first actuators facing each other across the second axis and having a piezoelectric element, and a pair of second actuators surrounding the first actuator, facing each other across the first axis, and having a piezoelectric element.

(14) It is preferable that the second actuator allows the mirror portion to swing around the first axis, and the first actuator allows the movable portion to swing around the second axis.

(15) It is preferable that the distance A and the total length L satisfy a relationship of $\frac{2}{3} < A/L < \frac{6}{5}$.

(16) An optical scanning device of the present disclosure comprises: the micromirror device according to any one of the aspects; and a processor that drives the driving portion, in which the processor allows the mirror portion to swing around the first axis and the second axis by providing a driving signal to the driving portion.

(17) According to the technique of the present disclosure, it is possible to provide a micromirror device and an optical scanning device capable of suppressing a shift of a resonance frequency by a crosstalk between axes.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Exemplary embodiments according to the technique of the present disclosure will be described in detail based on the following figures, wherein:
- (2) FIG. 1 is a schematic view of an optical scanning device,
- (3) FIG. 2 is a block diagram showing an example of a hardware configuration of a driving controller,
- (4) FIG. 3 is an external perspective view of a micromirror device,
- (5) FIG. 4 is a plan view of the micromirror device as viewed from a light incident side,
- (6) FIG. 5 is a cross-sectional view taken along the line A-A of FIG. 4,
- (7) FIG. 6 is a cross-sectional view showing a state where a mirror portion rotates around a first axis,
- (8) FIGS. 7A and 7B are diagrams showing examples of a first driving signal and a second driving signal,
- (9) FIG. 8 is a diagram schematically showing displacement of a second support portion in a case where the mirror portion rotates around the first axis,
- (10) FIG. 9 is a diagram showing parameters relating to dimensions of components of the micromirror device,
- (11) FIG. 10 is a diagram showing parameters relating to dimensions of components of the micromirror device,
- (12) FIG. 11 is a diagram showing specific setting values of the parameters,
- (13) FIG. 12 is a plan view of a micromirror device according to Modification Example,
- (14) FIG. 13 is a diagram showing parameters relating to dimensions of components of the micromirror device according to Modification Example,
- (15) FIG. 14 is a diagram showing parameters relating to dimensions of components of the micromirror device according to Modification Example,
- (16) FIG. 15 is a diagram showing parameters relating to dimensions of components of the micromirror device according to Modification Example,
- (17) FIG. 16 is a diagram showing specific setting values of the parameters,
- (18) FIG. 17 is a diagram showing experimental results for each sample,
- (19) FIG. 18 is a graph showing dependence of a shift amount of a resonance frequency on a ratio $A/Lb1$,
- (20) FIG. 19 is a graph showing dependence of power consumption on the ratio $A/Lb1$, and
- (21) FIG. 20 is a diagram showing simulation results.

DETAILED DESCRIPTION

- (22) An example of an embodiment relating to the technique of the present disclosure will be described with reference to the accompanying drawings.
- (23) FIG. 1 schematically shows an optical scanning device 10 according to an embodiment. The optical scanning device 10 includes a micromirror device (hereinafter, referred to as micromirror device (MMD)) 2, a light source 3, and a driving controller 4. The optical scanning device 10 optically scans a surface to be scanned 5 by reflecting a light beam L emitted from the light source 3 by the MMD 2 under the control of the driving controller 4. The surface to be scanned 5 is, for example, a screen.
- (24) The MMD 2 is a piezoelectric biaxial drive type micromirror device capable of allowing a mirror portion 20 (see FIG. 3) to swing around a first axis a.sub.1 and a second axis a.sub.2 orthogonal to the first axis a.sub.1. Hereinafter, the direction parallel to the first axis a.sub.1 is referred to as an X direction, the direction parallel to the second axis a.sub.2 is a Y direction, and the direction orthogonal to the first axis a.sub.1 and the second axis a.sub.2 is referred to as a Z

direction.

(25) The light source **3** is a laser device that emits, for example, laser light as the light beam L. It is preferable that the light source **3** emits the light beam L perpendicularly to a reflecting surface **20A** (see FIG. **3**) included in the mirror portion **20** in a state where the mirror portion **20** of the MMD **2** is stationary.

(26) The driving controller **4** outputs a driving signal to the light source **3** and the MMD **2** based on optical scanning information. The light source **3** generates the light beam L based on the input driving signal and emits the light beam L to the MMD **2**. The MMD **2** allows the mirror portion **20** to swing around the first axis a.sub.1 and the second axis a.sub.2 based on the input driving signal.

(27) As will be described in detail below, the driving controller **4** allows the mirror portion **20** to resonate around the first axis a.sub.1 and the second axis a.sub.2, so that the surface to be scanned **5** is scanned with the light beam L reflected by the mirror portion **20** such that a Lissajous waveform is drawn. This optical scanning method is called a Lissajous scanning method.

(28) The optical scanning device **10** is applied to, for example, a Lissajous scanning type laser display. Specifically, the optical scanning device **10** can be applied to a laser scanning display such as augmented reality (AR) glass or virtual reality (VR) glass.

(29) FIG. **2** shows an example of a hardware configuration of the driving controller **4**. The driving controller **4** has a central processing unit (CPU) **40**, a read only memory (ROM) **41**, a random access memory (RAM) **42**, a light source driver **43**, and an MMD driver **44**. The CPU **40** is an arithmetic unit that realizes the entire function of the driving controller **4** by reading out a program and data from a storage device such as the ROM **41** into the RAM **42** and executing processing. The CPU **40** is an example of a “processor” according to the technique of the present disclosure.

(30) The ROM **41** is a non-volatile storage device and stores a program for the CPU **40** to execute processing and data such as the optical scanning information described above. The RAM **42** is a volatile storage device that temporarily holds a program and data.

(31) The light source driver **43** is an electric circuit that outputs a driving signal to the light source **3** under the control of the CPU **40**. In the light source driver **43**, the driving signal is a driving voltage for controlling the irradiation timing and the irradiation intensity of the light source **3**.

(32) The MMD driver **44** is an electric circuit that outputs a driving signal to the MMD **2** under the control of the CPU **40**. In the MMD driver **44**, the driving signal is a driving voltage for controlling the timing, cycle, and deflection angle for allowing the mirror portion **20** of the MMD **2** to swing.

(33) The CPU **40** controls the light source driver **43** and the MMD driver **44** based on the optical scanning information. The optical scanning information is information including the scanning pattern of the light beam L with which the surface to be scanned **5** is scanned and the light emission timing of the light source **3**.

(34) Next, the configuration of the MMD **2** according to a first embodiment will be described with reference to FIGS. **3** to **5**. FIG. **3** is an external perspective view of the MMD **2**. FIG. **4** is a plan view of the MMD **2** as viewed from the light incident side. FIG. **5** is a cross-sectional view taken along the line A-A in FIG. **4**.

(35) As shown in FIG. **3**, the MMD **2** has the mirror portion **20**, a pair of first support portions **21**, a pair of movable frames **22**, a pair of second support portions **23**, a pair of first actuators **24**, a pair of second actuators **25**, a pair of first connecting portions **26A**, a pair of second connecting portions **26B**, and a fixed frame **27**. The MMD **2** is a so-called MEMS scanner.

(36) The mirror portion **20** has a reflecting surface **20A** for reflecting incident light. The reflecting surface **20A** is formed of a metal thin film such as gold (Au) and aluminum (Al) provided on one surface of the mirror portion **20**. The shape of the reflecting surface **20A** is, for example, circular with the intersection of the first axis a.sub.1 and the second axis a.sub.2 as the center.

(37) The first axis a.sub.1 and the second axis a.sub.2 exist, for example, in a plane including the reflecting surface **20A** in a case where the mirror portion **20** is stationary. The planar shape of the MMD **2** is rectangular, line-symmetrical with respect to the first axis a.sub.1, and line-symmetrical

with respect to the second axis a.sub.2.

(38) The pair of first support portions **21** are disposed at positions facing each other across the second axis a.sub.2, and have a shape that is line-symmetrical with respect to the second axis a.sub.2. In addition, each of the first support portions **21** has a shape that is line-symmetrical with respect to the first axis a.sub.1. The first support portion **21** is connected to the mirror portion **20** on the first axis a.sub.1, and swingably supports the mirror portion **20** around the first axis a.sub.1.

(39) The pair of movable frames **22** are disposed at positions facing each other across the first axis a.sub.1, and have a shape that is line-symmetrical with respect to the first axis a.sub.1. Each of the movable frames **22** has a shape that is line-symmetrical with respect to the second axis a.sub.2. In addition, each of the movable frames **22** is curved along the outer periphery of the mirror portion **20**. Both ends of the movable frame **22** are connected to the first support portion **21**.

(40) The first support portion **21** and the movable frame **22** are connected to each other to surround the mirror portion **20**. The mirror portion **20**, the first support portion **21**, and the movable frame **22** constitute the movable portion **60**.

(41) The pair of second support portions **23** are disposed at positions facing each other across the first axis a.sub.1, and have a shape that is line-symmetrical with respect to the first axis a.sub.1. Each of the second support portions **23** has a shape that is line-symmetrical with respect to the second axis a.sub.2. The second support portion **23** is connected to the movable frame **22** on the second axis a.sub.2, and swingably supports the movable portion **60** having the mirror portion **20** around the second axis a.sub.2. In addition, both ends of the second support portion **23** are connected to the first actuator **24**.

(42) The pair of first actuators **24** are disposed at positions facing each other across the second axis a.sub.2, and have a shape that is line-symmetrical with respect to the second axis a.sub.2. In addition, each of the first actuators **24** has a shape that is line-symmetrical with respect to the first axis a.sub.1. The first actuator **24** is formed along the outer periphery of the movable frame **22** and the first support portion **21**. The first actuator **24** is a piezoelectric drive type actuator comprising a piezoelectric element.

(43) In FIGS. **3** and **4**, although it seems that the first actuator **24** is divided in the vicinity of the first axis a.sub.1, the first actuator **24** is electrically connected by a wiring line (not shown) via the first axis a.sub.1.

(44) The second support portion **23** and the first actuator **24** are connected to each other to surround the movable portion **60**.

(45) The pair of second actuators **25** are disposed at positions facing each other across the first axis a.sub.1, and have a shape that is line-symmetrical with respect to the first axis a.sub.1. In addition, each of the second actuators **25** has a shape that is line-symmetrical with respect to the second axis a.sub.2. The second actuator **25** is formed along the outer periphery of the first actuator **24** and the second support portion **23**. The second actuator **25** is a piezoelectric drive type actuator comprising a piezoelectric element.

(46) In FIGS. **3** and **4**, although it seems that the second actuator **25** is divided in the vicinity of the second axis a.sub.2, the second actuator **25** is electrically connected by a wiring line (not shown) via the second axis a.sub.2.

(47) The pair of first connecting portions **26A** are disposed at positions facing each other across the second axis a.sub.2, and have a shape that is line-symmetrical with respect to the second axis a.sub.2. In addition, each of the first connecting portions **26A** has a shape that is line-symmetrical with respect to the first axis a.sub.1. The first connecting portion **26A** is disposed along the first axis a.sub.1, and connects the first actuator **24** and the second actuator **25** on the first axis a.sub.1.

(48) The pair of second connecting portions **26B** are disposed at positions facing each other across the first axis a.sub.1, and have a shape that is line-symmetrical with respect to the first axis a.sub.1. In addition, each of the second connecting portions **26B** has a shape that is line-symmetrical with respect to the second axis a.sub.2. The second connecting portion **26B** is disposed along the second

axis a.sub.2, and connects the second actuator **25** and the fixed frame **27** on the second axis a.sub.2.

(49) The second actuator **25** and the second connecting portion **26B** are connected to each other to surround the movable portion **60** and the first actuator **24**. The first actuator **24** and the second actuator **25** constitute a driving portion surrounding the movable portion **60**.

(50) The fixed frame **27** is a frame-shaped member having a rectangular outer shape, and has a shape that is line-symmetrical with respect to each of the first axis a.sub.1 and the second axis a.sub.2. The fixed frame **27** surrounds the outer periphery of the second actuator **25** and the second connecting portion **26B**. That is, the fixed frame **27** surrounds the driving portion.

(51) The first actuator **24** and the second actuator **25** are piezoelectric actuators each having a piezoelectric element. The pair of first actuators **24** allow the movable portion **60** to swing around the second axis a.sub.2 by applying rotational torque around the second axis a.sub.2 to the mirror portion **20** and the movable frame **22**. The pair of second actuators **25** allow the mirror portion **20** to swing around the first axis a.sub.1 by applying rotational torque around the first axis a.sub.1 to the mirror portion **20**, the movable frame **22**, and the first actuator **24**.

(52) As shown in FIG. **4**, the first support portion **21** is composed of a swing shaft **21A** and a pair of coupling portions **21B**. The swing shaft **21A** is a so-called torsion bar stretched along the first axis a.sub.1. One end of the swing shaft **21A** is connected to the mirror portion **20**, and the other end thereof is connected to the coupling portion **21B**.

(53) On the first axis a.sub.1, there is a spatial gap (hereinafter, referred to as a gap) **G1** between the first support portion **21** and the driving portion.

(54) The pair of coupling portions **21B** are disposed at positions facing each other across the first axis a.sub.1, and have a shape that is line-symmetrical with respect to the first axis a.sub.1. One end of the coupling portion **21B** is connected to the swing shaft **21A**, and the other end thereof is connected to the movable frame **22**. The coupling portion **21B** has a folded structure. Since the coupling portion **21B** has elasticity due to the folded structure, the internal stress applied to the swing shaft **21A** is relaxed in a case where the mirror portion **20** swings around the first axis a.sub.1.

(55) The second support portion **23** is composed of a swing shaft **23A** and a pair of coupling portions **23B**. The swing shaft **23A** is a so-called torsion bar stretched along the second axis a.sub.2. One end of the swing shaft **23A** is connected to the movable frame **22**, and the other end thereof is connected to the coupling portion **23B**.

(56) On the second axis a.sub.2, there is a gap **G2** between the second support portion **23** and the driving portion.

(57) The pair of coupling portions **23B** are disposed at positions facing each other across the second axis a.sub.2, and have a shape that is line-symmetrical with respect to the second axis a.sub.2. One end of the coupling portion **23B** is connected to the swing shaft **23A**, and the other end thereof is connected to the first actuator **24**. The coupling portion **23B** has a folded structure. Since the coupling portion **23B** has elasticity due to the folded structure, the internal stress applied to the swing shaft **23A** is relaxed in a case where the mirror portion **20** swings around the second axis a.sub.2.

(58) In the mirror portion **20**, a plurality of slits **20B** and **20C** are formed on the outside of the reflecting surface **20A** along the outer periphery of the reflecting surface **20A**. The plurality of slits **20B** and **20C** are disposed at positions that are line-symmetrical with respect to the first axis a.sub.1 and the second axis a.sub.2, respectively. The slit **20B** has an effect of suppressing distortion generated on the reflecting surface **20A** due to the swing of the mirror portion **20**.

(59) In FIGS. **3** and **4**, the wiring line and the electrode pad for giving the driving signal to the first actuator **24** and the second actuator **25** are not shown. A plurality of the electrode pads are provided on the fixed frame **27**.

(60) As shown in FIG. **5**, the MMD **2** is formed, for example, by performing an etching treatment on a silicon on insulator (SOI) substrate **30**. The SOI substrate **30** is a substrate in which a silicon

oxide layer **32** is provided on a first silicon active layer **31** made of single crystal silicon, and a second silicon active layer **33** made of single crystal silicon is provided on the silicon oxide layer **32**.

(61) The mirror portion **20**, the first support portion **21**, the movable frame **22**, the second support portion **23**, the first actuator **24**, the second actuator **25**, the first connecting portion **26A**, and the second connecting portion **26B** are formed of the second silicon active layer **33** remaining by removing the first silicon active layer **31** and the silicon oxide layer **32** from the SOI substrate **30** by an etching treatment. The second silicon active layer **33** functions as an elastic portion having elasticity. The fixed frame **27** is formed of three layers of the first silicon active layer **31**, the silicon oxide layer **32**, and the second silicon active layer **33**.

(62) The first actuator **24** includes a piezoelectric element (not shown) formed on the second silicon active layer **33**. The piezoelectric element has a laminated structure in which a lower electrode, a piezoelectric film, and an upper electrode are sequentially laminated on the second silicon active layer **33**. The second actuator **25** has the same configuration as the first actuator **24**.

(63) The upper electrode and the lower electrode are formed of, for example, gold (Au) or platinum (Pt). The piezoelectric film is formed of, for example, lead zirconate titanate (PZT), which is a piezoelectric material. The upper electrode and the lower electrode are electrically connected to the driving controller **4** described above via the wiring line and the electrode pad.

(64) A driving voltage is applied to the upper electrode from the driving controller **4**. The lower electrode is connected to the driving controller **4** via the wiring line and the electrode pad, and a reference potential (for example, a ground potential) is applied thereto.

(65) In a case where a positive or negative voltage is applied to the piezoelectric film in the polarization direction, deformation (for example, expansion and contraction) proportional to the applied voltage occurs. That is, the piezoelectric film exerts a so-called inverse piezoelectric effect. The piezoelectric film exerts an inverse piezoelectric effect by applying a driving voltage from the driving controller **4** to the upper electrode, and displaces the first actuator **24** and the second actuator **25**.

(66) FIG. **6** shows an example in which one piezoelectric film of the pair of second actuators **25** is extended and the other piezoelectric film is contracted, thereby generating rotational torque around the first axis a.sub.1 in the second actuator **25**. In this way, one of the pair of second actuators **25** and the other are displaced in opposite directions to each other, whereby the mirror portion **20** rotates around the first axis a.sub.1.

(67) In addition, FIG. **6** shows an example in which the second actuator **25** is driven in an anti-phase resonance mode (hereinafter, referred to as an anti-phase rotation mode) in which the displacement direction of the pair of second actuators **25** and the rotation direction of the mirror portion **20** are opposite to each other. On the other hand, an in-phase resonance mode in which the displacement direction of the pair of second actuators **25** and the rotation direction of the mirror portion **20** are the same direction is called an in-phase rotation mode. In the present embodiment, the second actuator **25** is driven in the anti-phase rotation mode.

(68) A deflection angle θ of the mirror portion **20** around the first axis a.sub.1 is controlled by the driving signal (hereinafter, referred to as a first driving signal) given to the second actuator **25** by the driving controller **4**. The first driving signal is, for example, a sinusoidal AC voltage. The first driving signal includes a driving voltage waveform $V_{\text{sub.1A}}(t)$ applied to one of the pair of second actuators **25** and a driving voltage waveform $V_{\text{sub.1B}}(t)$ applied to the other. The driving voltage waveform $V_{\text{sub.1A}}(t)$ and the driving voltage waveform $V_{\text{sub.1B}}(t)$ are in an anti-phase with each other (that is, the phase difference is 180°).

(69) The deflection angle θ of the mirror portion **20** around the first axis a.sub.1 corresponds to an angle at which the normal line N of the reflecting surface **20A** is inclined with respect to the Z direction in the YZ plane. Hereinafter, the deflection angle θ is also referred to as a rotation angle θ .

(70) The first actuator **24** is driven in an anti-phase resonance mode in the same manner as the

second actuator **25**. A deflection angle of the mirror portion **20** around the second axis a.sub.2 is controlled by the driving signal (hereinafter, referred to as a second driving signal) given to the first actuator **24** by the driving controller **4**. The second driving signal is, for example, a sinusoidal AC voltage. The second driving signal includes a driving voltage waveform V.sub.2A (t) applied to one of the pair of second actuators **25** and a driving voltage waveform V.sub.2B (t) applied to the other. The driving voltage waveform V.sub.2A (t) and the driving voltage waveform V.sub.2B (t) are in an anti-phase with each other (that is, the phase difference is 180°).

(71) FIGS. 7A and 7B show examples of the first driving signal and the second driving signal. FIG. 7A shows the driving voltage waveforms V.sub.1A (t) and V.sub.1B (t) included in the first driving signal. FIG. 7B shows the driving voltage waveforms V.sub.2A (t) and V.sub.2B (t) included in the second driving signal.

(72) The driving voltage waveforms V.sub.1A (t) and V.sub.1B (t) are represented as follows, respectively.

$$V_{\text{sub.1A}} = V_{\text{sub.off1}} + V_{\text{sub.1}} \sin(2\pi f_{\text{sub.d1}} t)$$

$$V_{\text{sub.1B}} = V_{\text{sub.off1}} + V_{\text{sub.1}} \sin(2\pi f_{\text{sub.d1}} t + \alpha)$$

(73) Here, V.sub.1 is the amplitude voltage. V.sub.off1 is the bias voltage. f.sub.d1 is the driving frequency (hereinafter, referred to as the first driving frequency). t is time. α is the phase difference between the driving voltage waveforms V.sub.1A (t) and V.sub.1B (t). In the present embodiment, for example, $\alpha = 180^\circ$.

(74) By applying the driving voltage waveforms V.sub.1A (t) and V.sub.1B (t) to the pair of second actuators **25**, the mirror portion **20** swings around the first axis a.sub.1 at the first driving frequency f.sub.d1.

(75) The driving voltage waveforms V.sub.2A (t) and V.sub.2B (t) are represented as follows, respectively.

$$V_{\text{sub.2A}} = V_{\text{sub.off2}} + V_{\text{sub.2}} \sin(2\pi f_{\text{sub.d2}} t + \varphi)$$

$$V_{\text{sub.2B}} = V_{\text{sub.off2}} + V_{\text{sub.2}} \sin(2\pi f_{\text{sub.d2}} t + \beta + \varphi)$$

(76) Here, V.sub.2 is the amplitude voltage. V.sub.off2 is the bias voltage. f.sub.d2 is the driving frequency (hereinafter, referred to as the second driving frequency). t is time. β is the phase difference between the driving voltage waveforms V.sub.2A (t) and V.sub.2B (t). In the present embodiment, for example, $\beta = 180^\circ$. In addition, φ is the phase difference between the driving voltage waveforms V.sub.1A (t) and V.sub.1B (t) and the driving voltage waveforms V.sub.2A (t) and V.sub.2B (t). In the present embodiment, for example, V.sub.off1 = V.sub.off2 = 0 V.

(77) By applying the driving voltage waveforms V.sub.2A (t) and V.sub.2B (t) to the pair of first actuators **24**, the movable portion **60** including the mirror portion **20** swings around the second axis a.sub.2 at the second driving frequency f.sub.d2.

(78) The first driving frequency f.sub.d1 is set so as to match the resonance frequency around the first axis a.sub.1 of the mirror portion **20**. The second driving frequency f.sub.d2 is set so as to match the resonance frequency around the second axis a.sub.2 of the mirror portion **20**. In the present embodiment, the first driving frequency f.sub.d1 is larger than the second driving frequency f.sub.d2.

(79) In a case where the MMD **2** configured as described above is driven two-dimensionally around the first axis a.sub.1 and the second axis a.sub.2, the centrifugal force acting in a case where the movable portion **60** swings around the second axis a.sub.2 serves to assist the swing of the mirror portion **20** around the first axis a.sub.1. The potential energy by the centrifugal force affects the total amount of elastic energy and kinetic energy stored in a case where the mirror portion **20** swings. As a result, a spring constant in the swing of the mirror portion **20** changes and a crosstalk between axes occurs, so that the resonance frequency shifts.

(80) The present applicant found that in a case where the displacement of the second support portion **23** satisfies a predetermined condition in a state where the mirror portion **20** rotates around the first axis a.sub.1 and an absolute value of the rotation angle θ is larger than 0 degrees, the shift

of the resonance frequency by the crosstalk between axes is suppressed.

(81) In a case where the displacement of the second support portion **23** satisfies a predetermined condition, a displacement amount of a portion of the movable portion **60** other than the mirror portion **20** increases. As a result, the total amount of elastic energy and kinetic energy in the entire MMD **2** is increased. As a result, the influence of the potential energy by the centrifugal force on the total energy is relatively reduced, and the shift of the resonance frequency by the crosstalk between axes is suppressed.

(82) FIG. **8** is a diagram schematically showing displacement of the second support portion **23** in a case where the mirror portion **20** rotates around the first axis a.sub.1. (A) of FIG. **8** shows a state where the mirror portion **20** is stationary and the rotation angle θ is 0 degrees. (B) of FIG. **8** shows a state where the mirror portion **20** rotates around the first axis a.sub.1 and the absolute value of the rotation angle θ is larger than 0 degrees.

(83) FIG. **8** shows a cross-sectional view of the MMD **2** cut along a plane orthogonal to the first axis a.sub.1 and including the second axis a.sub.2. A straight line α and an intersection C shown in (B) of FIG. **8** are included in the cross section of the MMD **2**. Specifically, the straight line α is a straight line located on the surface of the second support portion **23** and including each end point of the second support portion **23**. The intersection C is a point where the straight line α intersects the second axis a.sub.2.

(84) The total length of the second support portion **23** in the direction of the second axis a.sub.2 in the stationary state is denoted by Lb1. In addition, in the direction of the second axis a.sub.2, the distance from an end part of the second support portion **23** on the mirror portion **20** side in the stationary state to the intersection C is denoted by A. Through the experiment described below, the present applicant found that in a case where the distance A is larger than $\frac{2}{3}$ times the total length Lb1 (that is, in a case of $\frac{2}{3} \times Lb1 < A$), the shift of the resonance frequency by the crosstalk between axes is suppressed.

(85) In the experiment described below, the present applicant prepared a plurality of samples having different distances A for the MMD **2**, and measured the shift amount of the resonance frequency by the crosstalk between axes by driving each sample. Specifically, a resonance frequency around the first axis a.sub.1 in a case where each sample is driven one-dimensionally around the first axis a.sub.1 (hereinafter, referred to as a first resonance frequency fr1) and a resonance frequency around the first axis a.sub.1 in a case where each sample is driven two-dimensionally around the first axis a.sub.1 and the second axis a.sub.2 (hereinafter, referred to as a second resonance frequency fr2) were measured. Then, a shift amount Δfr of the resonance frequency by the crosstalk between axes was obtained by calculating a difference between the first resonance frequency fr1 and the second resonance frequency fr2.

(86) FIGS. **9** and **10** show parameters relating to the width, length, and the like of the components of the sample used in the experiment. FIG. **11** is a diagram showing specific setting values of the parameters.

(87) The diameter of the mirror portion **20** was 1.5 mm, the thickness of the SOI substrate **30** was 430 μm , the thickness of the second silicon active layer **33** was 60 μm , and the thickness of the silicon oxide layer **32** was 40 μm . The length of one side of the fixed frame **27** was 5.2 mm.

(88) The present applicant used Xac2 and Yac2 as variables among the parameters shown in FIG. **11**. That is, the present applicant prepared a plurality of samples having different distances A by changing the lengths of Xac2 and Yac2 for each sample.

Modification Example

(89) As Modification Example, the present applicant prepared samples for an MMD **2A** in which the shape and the like of the components are different from those of the MMD **2** according to the above embodiment.

(90) FIG. **12** shows the configuration of the MMD **2A** according to Modification Example. In FIG. **12**, the components having the same functions as those of the MMD **2** according to the above

embodiment are designated by the same reference numerals. In the MMD 2A, a connecting portion 26 is provided instead of the first connecting portion 26A and the second connecting portion 26B. The connecting portion 26 is provided on the first axis at, and connects the first actuator 24 to the second actuator 25 and connects the second actuator 25 to the fixed frame 27.

(91) FIGS. 13 to 15 show parameters relating to the width, length, and the like of the components of the MMD 2A. FIG. 16 is a diagram showing specific setting values of the parameters.

(92) In Modification Example, the diameter of the mirror portion 20 was 1.5 mm, the thickness of the SOI substrate 30 was 350 μm , the thickness of the second silicon active layer 33 is 60 μm , and the thickness of the silicon oxide layer 32 was 65 μm . The length of one side of the fixed frame 27 was 5.2 mm.

Experimental Result

(93) For the above-described embodiment and modification example, each sample was driven in a vacuum chamber to measure the first resonance frequency fr1 and the second resonance frequency fr2 . Specifically, the driving frequency was swept while irradiating the mirror portion 20 during driving with laser light, and the driving frequency at which a spreading angle of reflected light was maximized was measured as the resonance frequency. In addition, the deflection angle of the mirror portion 20 was calculated from the spreading angle of the reflected light.

(94) FIG. 17 shows measurement results of the first resonance frequency fr1 and the second resonance frequency fr2 for each sample. Sample number 1 indicates a sample prepared for the MMD 2A according to Modification Example. Sample numbers 2 to 9 indicate samples prepared for the MMD 2 according to the embodiment. Sample numbers 2 to 9 have different lengths of X_{ac2} and Y_{ac2} .

(95) The first resonance frequency fr1 is a resonance frequency during one-dimensional driving in a case where the deflection angle of the mirror portion 20 around the first axis $a_{\text{sub.1}}$ is ± 1.25 degrees. The second resonance frequency fr2 is a resonance frequency during two-dimensional driving in a case where the deflection angle of the mirror portion 20 around the first axis $a_{\text{sub.1}}$ is ± 1.25 degrees and the deflection angle of the mirror portion 20 around the second axis $a_{\text{sub.2}}$ is ± 11.5 degrees. The shift amount Δfr of the resonance frequency is a value obtained by subtracting the first resonance frequency fr1 from the second resonance frequency fr2 .

(96) The distance A was measured for each sample using a laser Doppler vibrometer. Then, using the measured distance A, a ratio $A/Lb1$ of the distance A to the total length $Lb1$ of the second support portion 23 was calculated.

(97) In a case where an application to a laser display for AR glasses is considered, an appropriate value of the deflection angle of the mirror portion 20 around each axis during the two-dimensional driving is ± 17 degrees around the first axis $a_{\text{sub.1}}$ and ± 11.5 degrees around the second axis $a_{\text{sub.2}}$. Therefore, for each sample, it was determined whether or not the two-dimensional driving was possible stably for 60 seconds or longer while maintaining the deflection angle at an appropriate value. The term “OK” indicates that the stable two-dimensional driving for 60 seconds or longer was possible. The term “NG” indicates that the stable two-dimensional driving for 60 seconds or longer was not possible.

(98) The power consumption of each sample was measured using a current probe in a state where the sample was resonated by the one-dimensional driving around the first axis $a_{\text{sub.1}}$. In this case, the deflection angle of the mirror portion 20 around the first axis $a_{\text{sub.1}}$ was ± 17 degrees.

(99) According to the experimental results shown in FIG. 17, it can be seen that the larger the ratio $A/Lb1$, the smaller the shift amount Δfr of the resonance frequency. The small shift amount Δfr means that an absolute value of the shift amount Δfr is small. In addition, it can be seen that in a case where the ratio $A/Lb1$ is larger than $\frac{2}{3}$ (greater than about 0.66), the two-dimensional driving is stabilized, and the superiority can be obtained from the viewpoint of the application to the laser display for AR glasses. FIG. 18 is a graph showing dependence of the shift amount Δfr of the resonance frequency on the ratio $A/Lb1$.

(100) FIG. 19 is a graph showing dependence of power consumption on the ratio $A/Lb1$. Low power consumption is desirable for a general laser display application. As a guide, in a case where the power consumption during the one-dimensional driving is 80 mW or less, a certain degree of superiority as a laser display application can be secured. Therefore, it can be seen that in a case where the ratio $A/Lb1$ is $6/5$ or less (1.2 or less), the power consumption is 80 mW or less, and a certain degree of superiority can be secured from the viewpoint of the power consumption.

(101) That is, it is preferable that the ratio $A/Lb1$ satisfies a relationship of $\frac{2}{3} < A/Lb1$. Further, it is preferable that the ratio $A/Lb1$ satisfies a relationship of $\frac{2}{3} < A/Lb1 < 6/5$.

(102) FIG. 20 shows the simulation results. (A) to (C) of FIG. 20 show that the lengths of $Xac2$ and $Yac2$ are different. The intersection C corresponds to a point where the displacement of the second support portion 23 is zero. It can be seen that the position of the intersection C changes (that is, the distance A changes) according to the lengths of $Xac2$ and $Yac2$. Also in the simulation, it was confirmed that the larger the ratio $A/Lb1$, the smaller the shift amount Δf_r of the resonance frequency.

(103) In the above embodiment, the hardware configuration of the driving controller 4 can be variously modified. The processing unit of the driving controller 4 may be composed of one processor or may be composed of a combination of two or more processors of the same type or different types. The processor includes, for example, a CPU, a programmable logic device (PLD), or a dedicated electric circuit. As is well known, the CPU is a general-purpose processor that executes software (program) to function as various processing units. The PLD is a processor such as a field programmable gate array (FPGA) whose circuit configuration can be changed after manufacture. The dedicated electric circuit is a processor that has a dedicated circuit configuration designed to perform a specific process, such as an application specific integrated circuit (ASIC).

(104) All documents, patent applications, and technical standards mentioned in this specification are incorporated herein by reference to the same extent as in a case where each document, each patent application, and each technical standard are specifically and individually described by being incorporated by reference.

Claims

1. A micromirror device comprising: a mirror portion on which a reflecting surface for reflecting incident light is formed; a pair of first support portions, each of which is connected to the mirror portion on a first axis located in a plane including the reflecting surface of the mirror portion in a stationary state, and swingably supports the mirror portion around the first axis; a pair of movable frames that are respectively connected to the pair of first support portions and face each other across the first axis; a pair of second support portions, each of which is connected to a corresponding one of the pair of movable frames on a second axis which is located in the plane including the reflecting surface of the mirror portion in the stationary state and is orthogonal to the first axis, and each of which includes a swing shaft that swingably supports a movable portion including the mirror portion, the first support portions, and the pair of movable frames around the second axis; a driving portion that surrounds the movable portion and has a gap with each of the pair of second support portions on the second axis; a pair of coupling portions included in each of the pair of second support portions, and configured to couple the swing shaft and the driving portion; and a fixed frame that is connected to the driving portion and surrounds the driving portion, wherein, in a state where the mirror portion rotates around the first axis and an absolute value of a rotation angle is larger than 0 degrees, assuming that, in a plane orthogonal to the first axis and including the second axis, a distance between an intersection between the second axis and a straight line located on a surface of the swing shaft and including each end point of the swing shaft and an end part of the swing shaft on a mirror portion side in the stationary state is denoted by A, and a total length of the swing shaft in a direction of the second axis is denoted by L, a

relationship of $\frac{2}{3} < A/L$ is satisfied.

2. The micromirror device according to claim 1, wherein the driving portion has a piezoelectric element.

3. The micromirror device according to claim 1, wherein the driving portion includes a pair of first actuators facing each other across the second axis and having a piezoelectric element, and a pair of second actuators surrounding the first actuator, facing each other across the first axis, and having a piezoelectric element.

4. The micromirror device according to claim 3, wherein the pair of second actuators allow the mirror portion to swing around the first axis, and the pair of first actuators allow the movable portion to swing around the second axis.

5. The micromirror device according to claim 1, wherein the distance A and the total length L satisfy a relationship of $\frac{2}{3} < A/L < 6/5$.

6. An optical scanning device comprising: the micromirror device according to claim 1; and a processor that drives the driving portion, wherein the processor allows the mirror portion to swing around the first axis and the second axis by providing a driving signal to the driving portion.
